

CoRoT-1b

R-2291

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_241121 · created 2026-08-17T00:56:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460635.9449 +/- 0.0051 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.027
Transit depth (Rp/Rs)^2	2.03 +/- 0.76 [%]
Orbital Inclination (inc)	85.99 +/- 2.59
Airmass coefficient 1 (a1)	0.987 +/- 0.028
Airmass coefficient 2 (a2)	0.015 +/- 0.077
Scatter in the residuals of the lightcurve fit is	2.85 %
Best Comparison Star	#1 - [541, 150]
Optimal Aperture	2.73
Optimal Annulus	5.97
Transit Duration (day)	0.106 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.143 $\pm$ 0.027 vs 0.1388 (deviation 0.12 σ)
Duration fit vs published	0.106 d vs 0.1046 d (deviation 0.14 σ)
Mid-time O-C	-1.8 min
Depth SNR	4.1
Residual scatter	2.85 %

Light curve

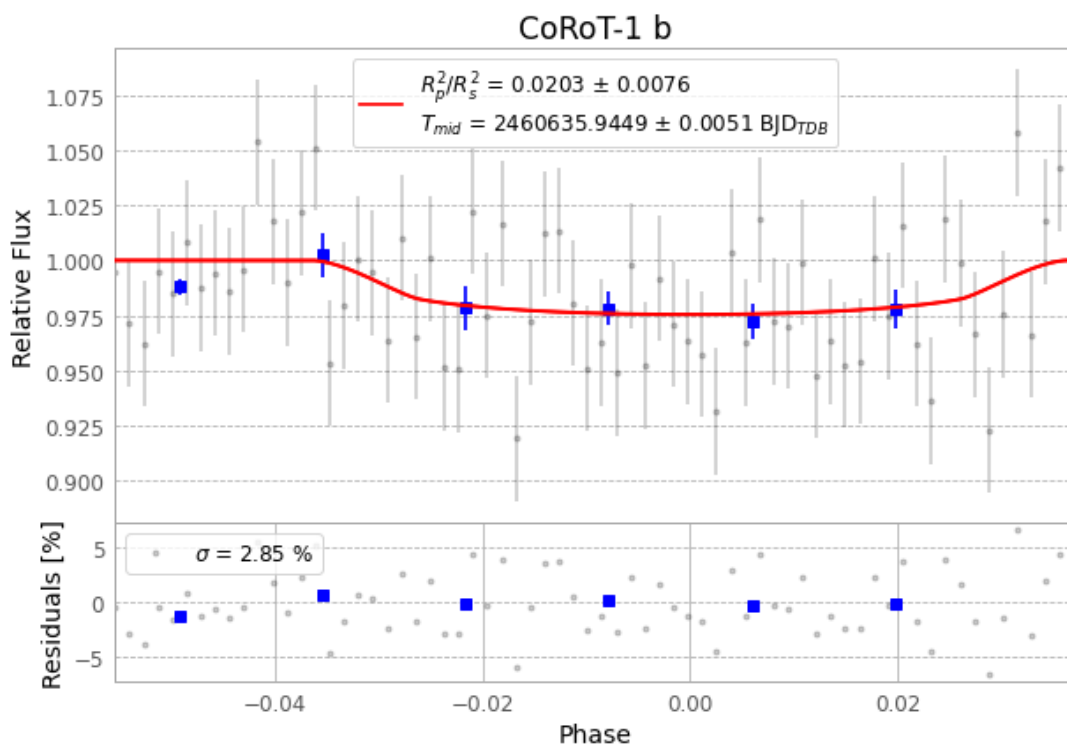


Figure 1 — FinalLightCurve_CoRoT-1 b_2024-11-21.png (SHA-256 cdd771b782dd84bc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 238], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 23586e27055d07a6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

68 FITS frames (45 MB), CoRoT-1241121083315.FITS → CoRoT-1241121115416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-241121.log (8ee4405f8a11...), FinalLightCurve_CoRoT-1 b_2024-11-21.png (cdd771b782dd...), FinalLightCurve_CoRoT-1 b_2024-11-21.csv (24529d8bfc93...)

Compute resources

Wall time 107.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 00:56Z.